

EC9540 - HUMAN COMPUTER INTERACTION

Lab 03

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MediLink: A Community Healthcare Information and Support Platform

Introduction

Healthcare services are essential for improving people's quality of life. However, many patients still face difficulties when accessing hospital services due to manual appointment booking, limited information about doctor and medicine availability and time-consuming payment processes. These challenges often lead to unnecessary hospital visits, long waiting times and inconvenience for patients.

To address these issues, we propose **MediLink**, a web-based healthcare platform designed to improve the patient experience through a simple and user-friendly interface. Based on the findings of our user requirements survey, the scope of the system focuses on three core patient services:

- Checking doctor availability and booking appointments.
- Checking medicine availability before visiting the hospital.
- Making secure online payments while providing patient details.

These features were selected because they directly address the main problems identified through the questionnaire survey. They also provide a complete and convenient user journey, allowing patients to check doctor availability, book appointments, verify medicine availability and complete payments online without unnecessary hospital visits. The system is designed following Human-Computer Interaction (HCI) principles to ensure usability, efficiency and a positive user experience.

Interface Design

Six core interfaces were planned for the MediLink patient platform, each mapped to a step in the HTA. Wireframes below use low-fidelity, greyscale/blue style deliberately, so the team can validate structure and flow before visual design is finalized. Every screen keeps the persistent top navigation bar, uses large touch-friendly buttons and high-contrast text (to satisfy the accessibility requirement for elderly users) and minimizes the number of fields/steps per screen (to satisfy the usability requirement identified from the 82.5% preference for “easy appointment booking”).

Login / Register Screen

- Single entry screen offering both Login and Register, since 100% of respondents own a smartphone but many are first-time platform users.
- Register panel supports a “registering on behalf of a dependent / elderly relative” checkbox, directly serving the Parents/Guardians and Elderly secondary user groups identified.
- Minimal required fields (email + password) to reduce drop-off before account verification.

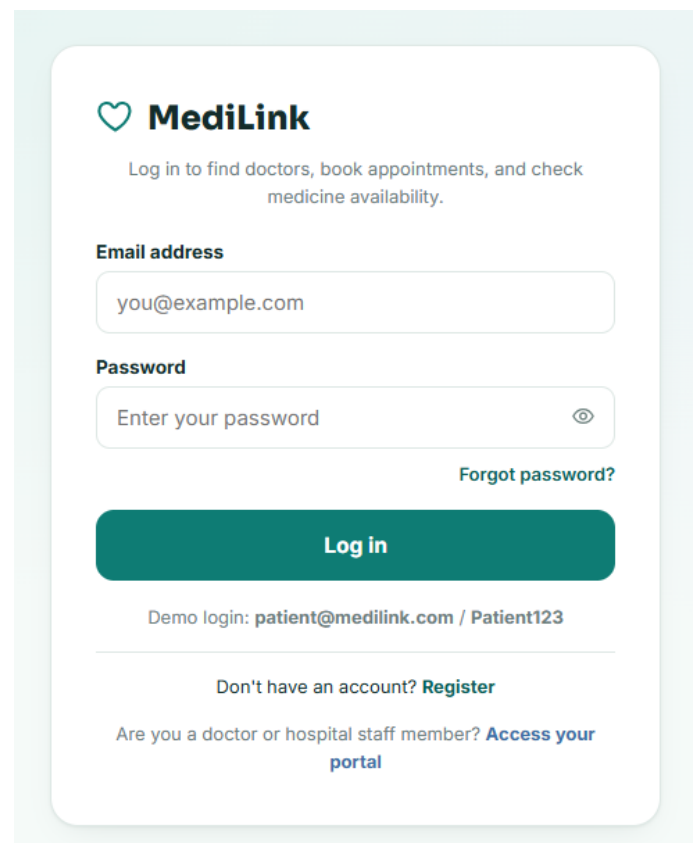
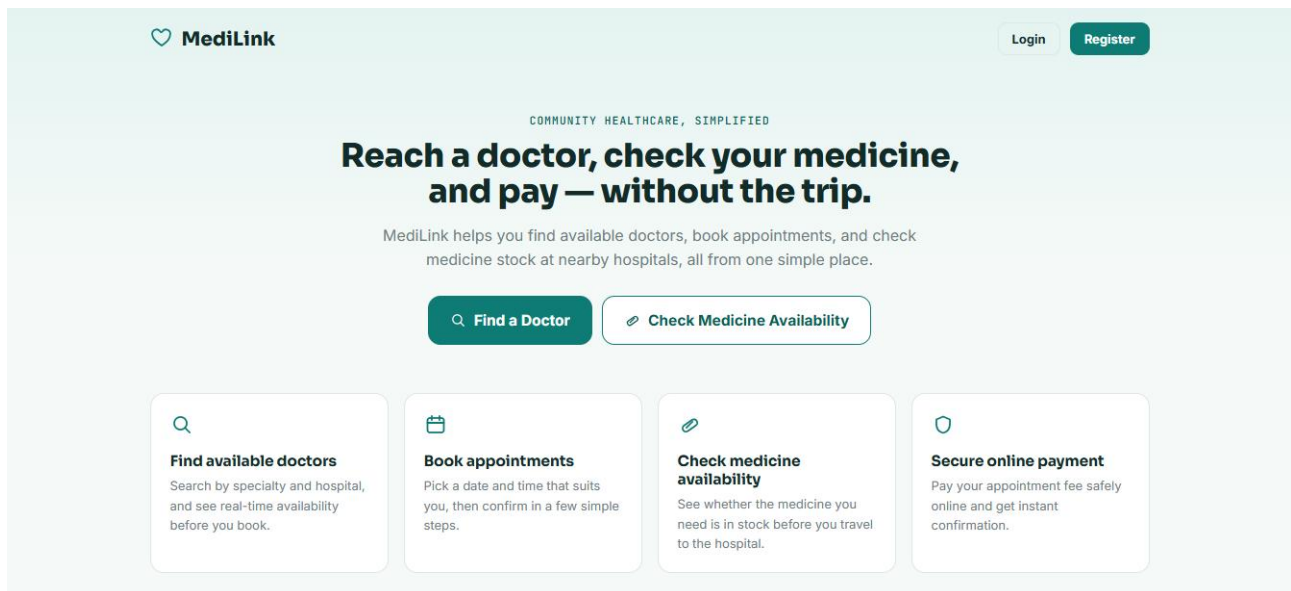


Figure 2.1: Wireframe - Login / Register Screen

Dashboard Screen

- Presents three quick action cards; Book an Appointment, Check Medicine Availability, My Appointments. So, the patient's next action is always one tap away.
- Shows the upcoming appointment status and recent medicine searches, reducing the need to re-navigate for information already retrieved.
- Acts as the hub screen in the navigation map.

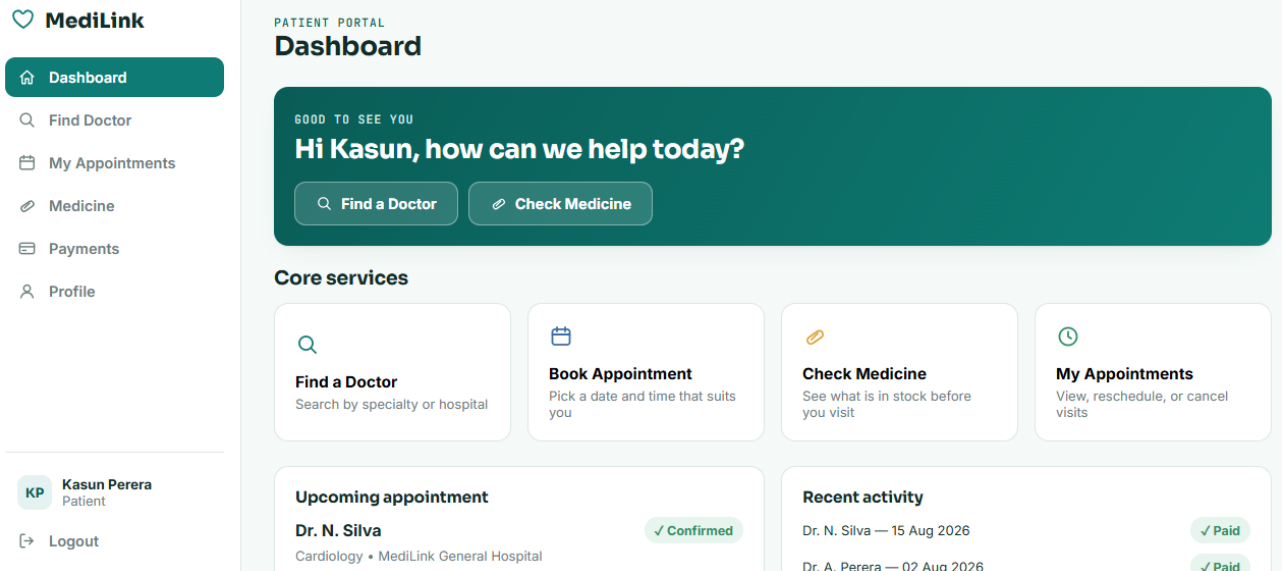


Figure 2.2: Wireframe - Dashboard Screen

Find Doctor & Book Appointment Screen

- Search/filter by specialty and hospital, addressing the 73.8% of respondents who requested “search doctors by specialty” and the 88.3% who requested “view available doctors.”
- Doctor’s result cards show live availability badges before the patient drills into time slots, avoiding unnecessary steps.
- An appointment summary panel confirms doctor, hospital and slot before the patient commits, reducing booking errors.

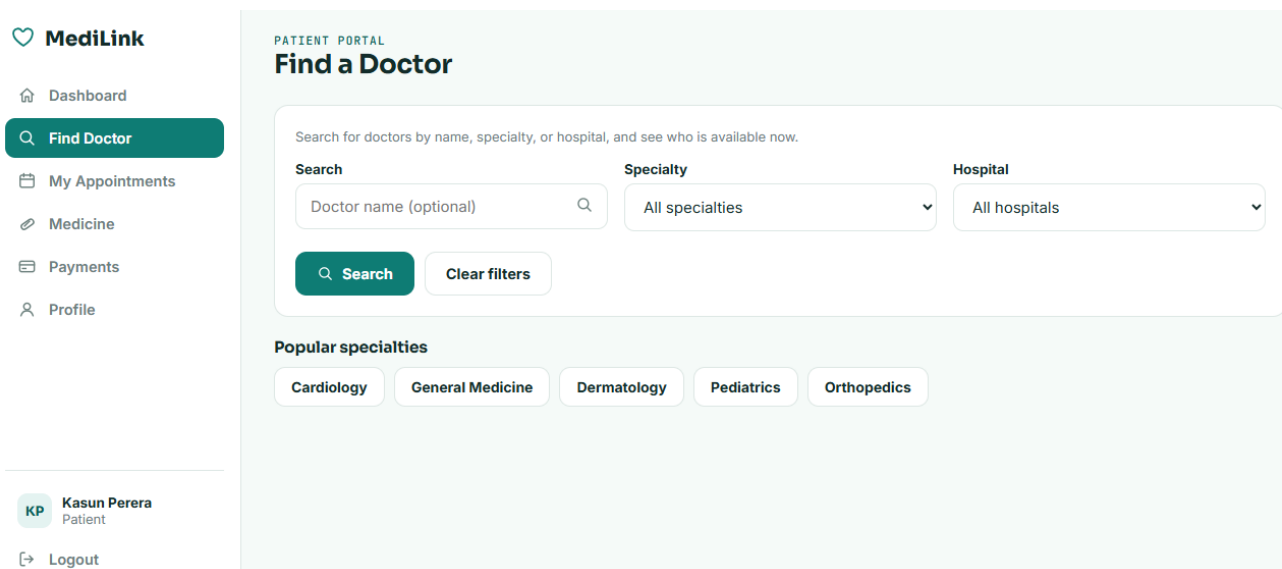


Figure 2.3: Wireframe - Find Doctor & Book Appointment Screen

Medicine Availability Screen

- Search by medicine name with an optional hospital filter, addressing the 92.2% of respondents who rated this feature useful and the 25.2% who had wasted a hospital trip on unavailable medicine.
- Results table shows stock status per hospital plus a last-updated timestamp, supporting the reliability of non-functional information requirement.
- “Book Anyway” and “Re-search” actions implement Plan 3 of the HTA, letting the patient decide how to proceed based on the result.

The wireframe shows the 'Medicine Availability' screen within the 'MediLink' Patient Portal. On the left is a sidebar with navigation links: Dashboard, Find Doctor, My Appointments, Medicine (highlighted), Payments, and Profile. The main content area has a header 'PATIENT PORTAL Medicine Availability'. Below the header is a message: 'Check medicine availability before visiting the hospital.' The search form consists of two input fields: 'Medicine name' (with a placeholder 'e.g. Paracetamol') and 'Hospital / pharmacy' (with a dropdown menu showing 'All Hospitals'). A green 'Search' button is to the right of the hospital field. At the bottom left, a user profile for 'Kasun Perera' (Patient) is shown with a 'Logout' link.

Figure 2.4: Wireframe - Medicine Availability Screen

Patient Details & Payment Screen

- Combines patient detail entry, fee summary and payment method selection on one screen, consistent with the 64.1% of respondents who rated secure payment important and the preference for minimal steps.
- A visible security notice reassures patients that the payment gateway is encrypted, addressing the security requirement.
- Radio-button payment method selection keeps the common case (card payment) visible by default.

< Back

PATIENT PORTAL

Payment

Doctor Slot Summary Details **Payment** Done

Payment Details

You are paying to
MediLink General Hospital

Appointment with **Dr. N. Silva** Date & Time **20 Aug 2025, 10:00 AM**

Amount to Pay
Rs. 3,500.00

Select Payment Method

☒ **Credit / Debit Card**
Visa, MasterCard, American Express

☐ **Other Methods**
UPI / Internet Banking / Wallets

Pay with Card

Accepted Cards

VISA MasterCard AMEX

Card Number

1234 5678 9012 3456

Expiry Date

MM / YY

CVV

123

Cardholder Name

Name as on card

Your payment information is secure and encrypted.

Pay Rs. 3,500.00

Figure 2.5: Wireframe - Patient Details & Payment Screen

Booking Confirmation Screen

- Provides the on-screen confirmation required by the functional requirements, with a booking reference, appointment summary and payment status.
- Offers a direct shortcut back into My Appointments for record-keeping.
- A downloadable receipt supports the patient's need for proof of payment.

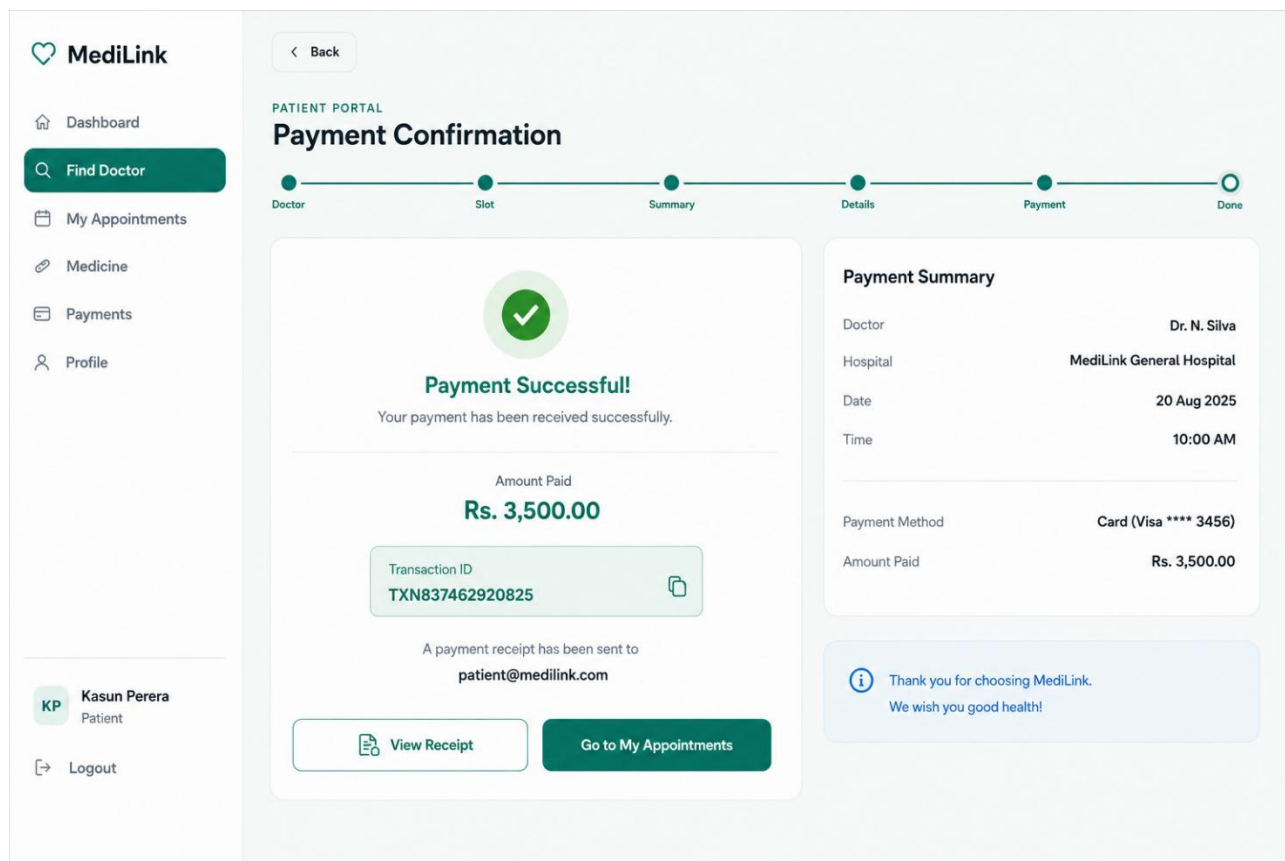


Figure 2.6: Wireframe - Booking Confirmation Screen

Menus

MediLink uses a single persistent left-navigation menu for the authenticated patient area, supplemented by inline filter menus (dropdowns) on the Find Doctor and Medicine Check screens and a profile drop down menu. Keeping menu depth to a maximum of two levels satisfies the requirement to minimize steps for both younger and elderly users.

Primary Navigation Menu

Visible on every screen once the patient is logged in:

Menu Item	Destination	Purpose
Dashboard	Dashboard screen	Return to the patient home / quick actions
Find Doctor	Find Doctor & Book Appointment screen	Find a doctor
Medicine Check	Medicine Availability screen	Check medicine availability
My Appointments	Appointment list screen	View, reschedule or cancel bookings

Menu Item	Destination	Purpose
Profile	Profile drop-down menu	Account-related actions
Logout	Landing / Login screen	End the session securely

Contextual Filter Menus

- Find Doctor screen: Specialty drop-down (e.g., Cardiology, General Physician) and Hospital drop-down, both populated from back-end reference data.
- Medicine Check screen: Hospital/Pharmacy drop-down, restricting the stock search to a chosen location.
- My Appointments screen: Status filter menu (Upcoming / Completed / Cancelled).

Profile Drop-Down Menu

- Editing Personal Details
- Manage Dependents (for guardians booking on behalf of family members)
- Change Password
- Notification Preferences (SMS / email)
- Logout

Secondary Portals (Doctor & Hospital Staff)

The current build scope is patient-only. For completeness, the doctor and hospital-staff roles identified as secondary users would use separate, role-restricted menus: Doctors would see “My Schedule” and “Today's Appointments;” Hospital Staff would see “Manage Doctors,” “Manage Medicine Stock” and “Verify Payments.” These are noted here for planning continuity but are out of scope for the current patient-platform build.

GUI Elements

The table below catalogues the interface controls required across the platform, drawn directly from the wireframes, together with where each is used and why.

Element	Used On	Purpose
Text field	Login/Register, Find Doctor, Medicine Check, Payment	Free-text entry: credentials, names, contact details, search terms, card details
Dropdown / select	Find Doctor, Medicine Check, My Appointments	Constrained choice: specialty, hospital, status filter

Element	Used On	Purpose
Primary button	All screens	Main action per screen (Login, Search, Confirm Appointment, Pay & Confirm)
Secondary (outline) button	Login, Medicine Check, Confirmation	Alternative / less-critical action (Register, Book Anyway, Download Receipt)
Checkbox	Register panel	Optional flags, e.g. registering on behalf of a dependent
Radio button	Payment screen	Mutually exclusive choice of payment method
Data table	Dashboard, Medicine Check	Tabular results: medicine stock, recent searches
Card component	Dashboard, Find Doctor results	Grouped summary of an item (quick action, doctor)
Status badge / tag	Dashboard, Find Doctor, Medicine Check	At-a-glance status: Confirmed, In Stock, Out of Stock
Slot / chip selector	Find Doctor – time slots	Single-select list of discrete options (appointment times)
Top navigation bar	All authenticated screens	Primary menu access (Section 3.1)
Icon / avatar placeholder	Dashboard, Find Doctor	Visual identification for doctors and quick actions
Confirmation banner / summary panel	Book Appointment, Payment, Confirmation	Reinforces key details before/after a committing action
Security notice	Payment screen	Communicates that the transaction is protected (non-functional security requirement)

Navigation

The navigation model follows the natural sub-task sequence. While still allowing the patient to jump directly to any screen via the persistent top menu. The diagram below shows both the primary linear path (solid arrows) and the shortcut / alternate paths (dashed arrows).

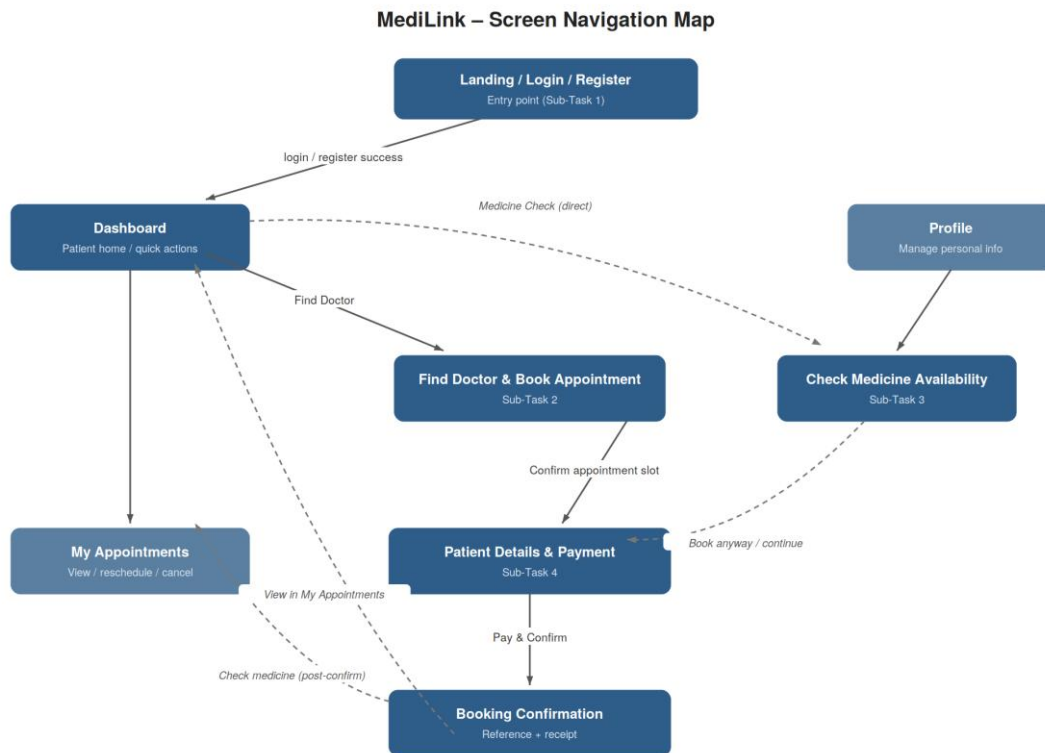


Figure 5.1: MediLink screen navigation map

Primary Path (End-to-End Booking Journey)

- Landing / Login / Register → Dashboard (on successful login)
- Dashboard → Find Doctor & Book Appointment (Find Doctor selected)
- Find Doctor & Book Appointment → Patient Details & Payment (slot confirmed)
- Patient Details & Payment → Booking Confirmation (payment successful)

Shortcuts and Alternate Paths

- Dashboard or Confirmation → Medicine Availability directly, for patients who only need to check stock.
- Medicine Availability → Patient Details & Payment (“Book Anyway”), re-joining the primary path without losing context.
- Confirmation → My Appointments, so the new booking is immediately visible in the patient's list.
- Any screen → any top-level menu item, via the persistent navigation bar, so the model is a network rather than a strict wizard; patients are never trapped in a linear flow.

Back-End Plan

Data Requirements

Based on the functional requirements, the back end must persist and serve the following categories of data:

- Patient accounts and dependent/guardian relationships
- Doctor profiles, specialties, hospital affiliation, and availability slots
- Hospital and hospital-staff reference data
- Appointment records and their lifecycle status (pending payment, confirmed, completed, cancelled, rescheduled)
- Medicine catalogue and per-hospital stock levels with last-updated timestamps
- Payments, transaction status, and receipts
- Notification logs (SMS/email confirmations)

Proposed Database Schema

A relational database is proposed since the data is highly structured and relationships (patient-appointment-doctor-payment) require referential integrity. Key tables:

Table	Key Fields	Notes
patient	patient_id (PK), name, nic, mobile, email, password_hash, dob, guardian_of	guardian_of self-references patient_id for dependants
doctor	doctor_id (PK), name, specialty, hospital_id (FK), contact	Linked to hospital
hospital	hospital_id (PK), name, address, contact	Reference data managed by hospital staff
hospital_staff	staff_id (PK), name, role, hospital_id (FK)	Manages doctor/medicine reference data
doctor_availability	availability_id (PK), doctor_id (FK), date, start_time, end_time, status	status: open / booked
appointment	appointment_id (PK), patient_id (FK), doctor_id (FK), availability_id (FK), status, created_at	status: pending_payment / confirmed / completed / cancelled
medicine	medicine_id (PK), name, category	Medicine catalogue

Table	Key Fields	Notes
medicine_stock	stock_id (PK), medicine_id (FK), hospital_id (FK), quantity, status, last_updated	status: in_stock / out_of_stock
payment	payment_id (PK), appointment_id (FK), amount, method, status, transaction_ref, paid_at	status: pending / successful / failed
receipt	receipt_id (PK), payment_id (FK), receipt_number, generated_at	Downloadable proof of payment
notification	notification_id (PK), patient_id (FK), type, channel, message, sent_at	type: booking_confirmation / reminder

Major Back-End Functions

Grouped by service, matching the application-tier services shown in the system model:

Auth Service

- registerPatient(details) - creates a new patient account, including guardian/dependant linkage
- loginPatient(credentials) - authenticates and issues a session/token
- verifyAccount(otp) - confirms mobile/email ownership
- resetPassword(request) - secure password recovery flow

Doctor & Appointment Service

- getDoctors(specialty, hospital) - returns filtered doctor list with next-available slot
- getDoctorAvailability(doctorId) - returns open time slots
- bookAppointment(patientId, availabilityId) - creates a pending_payment appointment and locks the slot
- rescheduleAppointment(appointmentId, newSlotId) / cancelAppointment(appointmentId)
- getPatientAppointments(patientId) - returns the patient's appointment history and status

Medicine Availability Service

- searchMedicine(name, hospitalId) - returns stock status and last-updated time
- updateMedicineStock(staffId, medicineId, hospitalId, quantity) - restricted to hospital staff

Payment Service

- calculateFee(appointmentId) - derives the consultation fee
- initiatePayment(appointmentId, method) - begins a transaction with the payment gateway
- confirmPayment(transactionRef) - verifies gateway callback and updates appointment/payment status
- generateReceipt(paymentId) - produces a downloadable receipt

Notification Service

- `sendConfirmation(appointmentId)` - SMS/email after successful booking and payment
- `sendReminder(appointmentId)` - scheduled reminder ahead of the appointment

Non-Functional Back-End Considerations

- **Performance:** doctor and medicine search queries should be indexed on specialty/hospital and medicine name/hospital respectively, to keep result pages fast (43.7% of respondents rated fast loading important).
- **Reliability:** `medicine_stock.last_updated` must be surfaced to the patient and refreshed by hospital staff regularly, since the value of this feature depends on currency of data.
- **Security:** `password_hash` storage (never plain text), HTTPS everywhere, and payment details handled via a PCI-compliant gateway rather than stored directly in the MediLink database.
- **Concurrency:** `doctor_availability` slot locking is required at booking time to prevent two patients from double-booking the same slot.

Overall System Model

MediLink is planned as a three-tier web architecture: a responsive front-end client, a back-end application server exposing and a data tier combining the relational database with the external payment gateway. This separation lets the front-end evolve independently of business logic and lets business logic scale independently of the presentation layer.

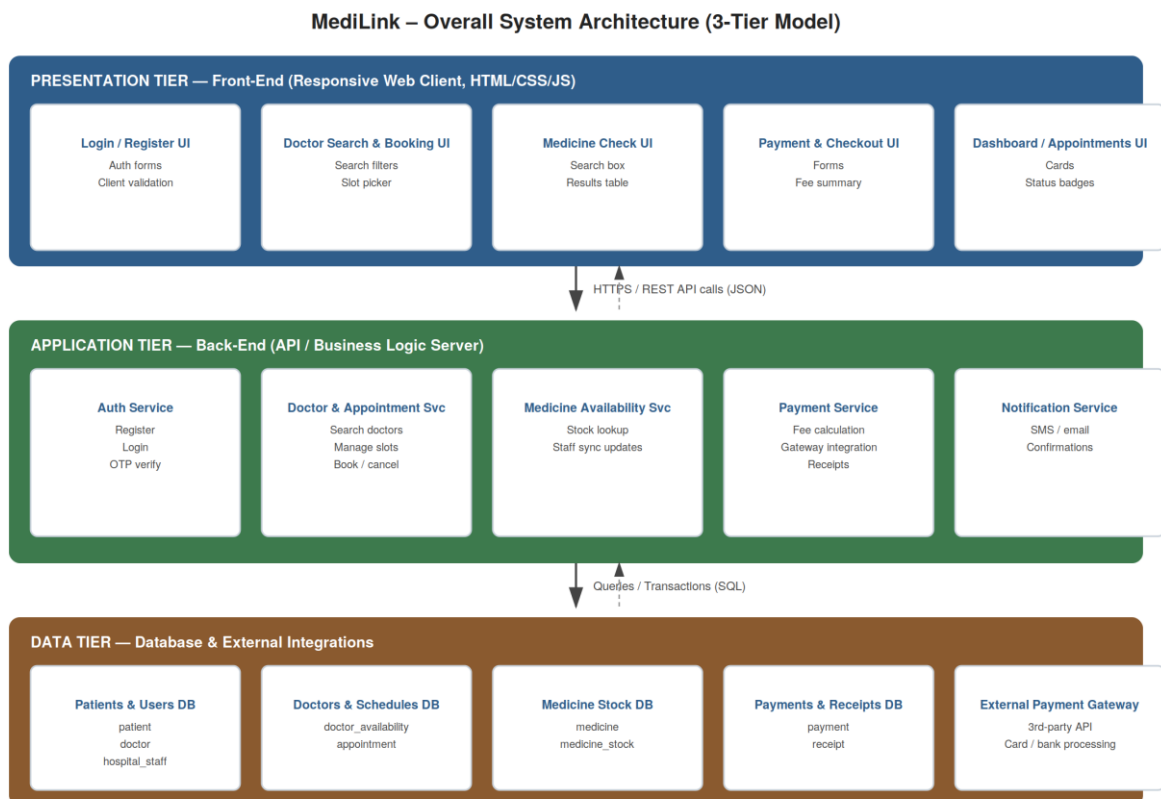


Figure 7.1: MediLink overall system architecture

How the Tiers Connect

- The front-end client calls the back end exclusively via HTTPS REST endpoints returning JSON, so the same API can later serve a mobile app without change.
- The back-end application tier is organized into five focused services (Auth, Doctor & Appointment, Medicine Availability, Payment, Notification) each owning its own functions but sharing the same database.
- The data tier holds the relational database and integrates with an external payment gateway for actual transaction processing, keeping card data out of the MediLink database entirely.
- Responses flow back up the same path: database → application service → API response → front-end UI update, keeping the patient's on-screen state consistent with the stored data.

Conclusion

This planning report has translated, buildable design: six wireframed interfaces covering the full patient journey, a shallow two-level menu structure, a catalogued set of reusable GUI elements, a navigation model that supports both the primary booking journey and flexible shortcuts, a relational back-end plan with clearly scoped services and functions, and a three-tier system architecture connecting it all together.

Every design decision traces back to a specific requirement or questionnaire finding from Lab 02, ensuring the team builds exactly what the 103 surveyed users asked for easy doctor search and booking, reliable medicine availability checking and secure, convenient online payment without scope creep. This report is intended to directly guide the next stage of the project: interactive prototyping and development.